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A Minimum Squared-Error Framework for Sampling and Reconstruction in Arbitrary Spaces

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Abstract

We consider non-ideal sampling and reconstruction schemes in which the sampling and reconstruction spaces as well as the input signal can be arbitrary. To obtain a good reconstruction of the signal in the reconstruction space from arbitrary samples, we suggest processing the samples prior to reconstruction with a linear transformation that is designed to minimize the worst-case squared-norm error between the reconstructed signal, and the best possible (but usually unattainable) approximation of the signal in the reconstruction space. If the input signal is known to lie in an appropriately chosen subspace, then we propose a linear transformation that achieves the minimal squared-error norm approximation. We show both theoretically and through simulations that if the input signal does not lie in the reconstruction space, then the suggested methods can outperform the consistent reconstruction method previously proposed for this problem.

1 Introduction

The most common setting considered in the sampling literature is that introduced by Shannon's sampling theorem, in which the input signal is assumed to be bandlimited, the samples of the signal are ideal, *i.e.,* they are equal to the signal values at a set of sampling points, and the reconstructed signal is also a bandlimited function, created by using the sinc interpolation kernel. In practice, however, the input signal is never perfectly bandlimited, and the sampling process is never ideal. Another drawback of the Shannon paradigm is the difficulty in implementing the infinite sinc interpolating kernel, which has slow decay.

To overcome these limitations of the traditional sampling framework, a more recent approach to sampling is to consider samples that can be represented as the inner products of the input signal x with a set of sampling vectors, which form a possibly overcomplete basis (frame) for the sampling space, denoted by $\mathcal{S}$ [1, 2, 3, 4, 5, 6, 7, 8, 9, 10]. Examples include multiresolution decompositions [2, 11], and spline

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decompositions [4]. The problem then is to reconstruct x from these samples, using a set of vectors that form an overcomplete bases for a subspace $\mathcal{W}$, referred to as the reconstruction space. The input signal x is assumed to lie in an arbitrary Hilbert space $\mathcal{H}$. Since, in this framework, the reconstructed signal is constrained to lie in $\mathcal{W}$, if x does not lie in $\mathcal{W}$, then perfect reconstruction is not possible, regardless of the sampling and reconstruction method.

If the sampling space $\mathcal{S}$ is equal to the reconstruction space $\mathcal{W}$, then by properly processing the samples prior to reconstruction we can obtain the minimal-error approximation of x in the space $\mathcal{W}$, given by $P_{\mathcal{W}}x$, where $P_{\mathcal{A}}$ denotes the orthogonal projection onto the subspace $\mathcal{A}$. Indeed, the signal $\hat{x} = P_{\mathcal{W}}x$ minimizes the squared-norm error between x and any $\hat{x}$ in $\mathcal{W}$. However, if $\mathcal{S}$ is arbitrary, then the minimal-norm approximation is usually unattainable. Therefore, our problem is to design a linear transformation H to process the given samples prior to reconstruction (see Fig. 1), so that the reconstructed signal $\hat{x}$ is close to x in a squared-error sense.

The problem depicted in Fig. 1 was first considered in [1], for the case in which the sampling and reconstruction spaces are shift-invariant (SI) subspaces of L_2, *i.e.,* spaces generated by translates of an appropriately chosen function. The transformation H was chosen such that $\hat{x}$ is a *consistent reconstruction* of x, namely a reconstruction with the property that it yields the same samples as x. This approach was then generalized in [7, 10, 12] to arbitrary sampling and reconstruction spaces, as well as arbitrary input spaces $\mathcal{H}$. The resulting reconstruction is $\hat{x} = E_{\mathcal{W}\mathcal{S}^\perp}x$ where $E_{\mathcal{W}\mathcal{S}^\perp}$ is the oblique projection onto $\mathcal{W}$ along $\mathcal{S}^\perp$, and $\mathcal{S}^\perp$ is the orthogonal complement of $\mathcal{S}$. Note, however, that the fact that x and $\hat{x}$ yield the same samples does not necessarily imply that $\hat{x}$ is close to x in the squared-norm sense. In fact, the norm of the resulting reconstruction error $\hat{x} - x$ can be made arbitrarily large, if $\mathcal{S}$ is close to $\mathcal{W}^\perp$.

To obtain a reconstruction $\hat{x}$ that is close to x, we may choose H to minimize the squared-norm error $\|\hat{x} - x\|^2$. However, since this error depends on x, which is unknown, it cannot generally be minimized directly. An exception is the case in which x is known to lie in a subspace $\mathcal{A}$ of $\mathcal{H}$, not necessarily equal to $\mathcal{W}$, that satisfies a certain decomposition property. In this case, we show in Section 4, that the minimal norm reconstruction is attainable.

In the general case where x can be an arbitrary vector in $\mathcal{H}$, instead of minimizing the squared-error, we may consider a worst-case approach in which we design H to minimize the worst-case error norm over all bounded energy inputs x. The drawback of such a minimax approach is that it may tend to be over-conservative, since it considers the worst possible input choice. Indeed, as we will show in Section 3, the minimax transformation is given by $H = 0$. To obtain a reconstruction that is less pessimistic, we consider a competitive approach, similar in spirit to the approaches in [13, 14], in which we seek the transformation H that minimizes the worst-case *regret* instead of the worst-case error norm, where we define the regret as

the difference between the error norm achieved by a transformation H, and the smallest possible error norm achievable by a transformation H that depends on the unknown input x. As we show, the minimax regret reconstruction is given by $\hat{x} = P_{\mathcal{W}} P_{\mathcal{S}} x$. This reconstruction has the desirable property that the resulting reconstruction error is always bounded by twice the norm of x. This is in contrast with the consistent reconstruction method which can result in an arbitrarily large reconstruction error.

When the spaces $\mathcal{S}$ and $\mathcal{W}$ are SI subspaces of L_2, then we show in Section 5 that the minimax regret reconstruction and the minimal norm reconstruction, when x is constrained to an appropriate subspace, can be implemented using linear time-invariant (LTI) discrete-time filters. A particularly efficient implementation of these filters is possible in the case of spline spaces, based on the results of [15, 16, 4, 17].

To compare the minimax regret method with the consistent reconstruction method, we develop bounds on the reconstruction error resulting from both approaches. In particular, we show that if the spaces $\mathcal{S}$ and $\mathcal{W}$ are far enough apart, or if x has enough energy in $\mathcal{S}$, then the worst-case and best-case error using the minimax regret reconstruction are smaller than the worst-case and best-case error using the consistent reconstruction. These theoretical results are also demonstrated through simulations in Section 7.

The outline of the paper is as follows. In Section 2 we introduce the sampling and reconstruction problem and show that the minimal squared-error reconstruction cannot be obtained in general with a linear transformation. In Section 3 we develop two minimax approaches to sampling and reconstruction: the minimax squared-error approach and the minimax regret approach. Section 4 considers minimization of the squared-norm error over a subspace of $\mathcal{H}$. The special case of sampling and reconstruction in SI subspaces is considered in Section 5. In Section 6 we analyze the reconstruction error resulting from the minimax regret approach, and compare it with the error resulting from consistent reconstruction. Simulation results are presented in Section 7.

2 Problem Formulation and Main Results

2.1 Notation

We denote vectors in an arbitrary Hilbert space $\mathcal{H}$ by lowercase letters, and the elements of a sequence $c \in \ell_2$ by $c[n]$. The orthogonal projection operator onto a subspace $\mathcal{A}$ of $\mathcal{H}$ is denoted by $P_{\mathcal{A}}$, the orthogonal complement of $\mathcal{A}$ is denoted by $\mathcal{A}^{\perp}$, and $\mathcal{N}(\cdot)$ and $\mathcal{R}(\cdot)$ denote the null space and range space of the corresponding operator, respectively. The *Moore-Penrose pseudo inverse* [18] of a transformation T is denoted by $T^{\dagger}$, and T^* denotes the adjoint of T. The inner product between vectors $x, y \in \mathcal{H}$ is denoted by $\langle x, y \rangle$, and is linear in the second argument.

A set transformation $V : \ell_2 \to \mathcal{H}$ corresponding to vectors $\{v_n\} \in \mathcal{H}$ is defined by $Va = \sum_n a[n] v_n$ for

all $a \in \ell_2$. From the definition of the adjoint, if $a = V^* y$, then $a[n] = \langle v_n, y \rangle$.

We denote by $E_{\mathcal{WS}^\perp}$ the *oblique projection* [1][19, 20, 21] onto $\mathcal{W}$ along $\mathcal{S}^\perp$, so that $\mathcal{R}(E_{\mathcal{WS}^\perp}) = \mathcal{W}$ and $\mathcal{N}(E_{\mathcal{WS}^\perp}) = \mathcal{S}^\perp$. The oblique projection $E_{\mathcal{WS}^\perp}$ is defined as the unique operator satisfying

$$\begin{aligned} E_{\mathcal{WS}^\perp} w &= w \text{ for any } w \in \mathcal{W}; \\ E_{\mathcal{WS}^\perp} v &= 0 \text{ for any } v \in \mathcal{S}^\perp. \end{aligned} \tag{1}$$

The operator $E_{\mathcal{WS}^\perp}$ can be expressed as [12],[22]

$$E_{\mathcal{WS}^\perp} = W \left(S^* W\right)^\dagger S^* \tag{2}$$

where S and W are bounded operators with $\mathcal{R}(S) = \mathcal{S}$ and $\mathcal{R}(W) = \mathcal{W}$. As a special case, the orthogonal projection $P_{\mathcal{W}}$ can be written as

$$P_{\mathcal{W}} = W \left(W^* W\right)^\dagger W^*. \tag{3}$$

2.2 Problem Formulation

Suppose we are given measurements $c[n]$ of a signal x that lies in an arbitrary Hilbert space $\mathcal{H}$. The measurements are obtained by taking the inner product of the signal with a set of sampling vectors $\{s_n, n \in \mathcal{I}_s\}$, where $\mathcal{I}_s$ is some index set, so that $c[n] = \langle s_n, x \rangle$. We construct an approximation $\hat{x}$ of x using a given set of reconstruction vectors $\{w_n, n \in \mathcal{I}_w\}$, where $\mathcal{I}_w$ is an index set, so that the reconstruction $\hat{x}$ has the form

$$\hat{x} = \sum_n d[n] w_n \tag{4}$$

for some coefficients $d[n]$ that are a linear transformation of the samples $c[n]$. Denoting by d and c the sequences in ℓ_2 with elements $d[n]$ and $c[n]$ respectively, we have that $d = Hc$ for some linear transformation H. Our objective is to derive a linear transformation H such that the reconstructed signal $\hat{x}$ will be close to the original signal x in the squared-norm sense.

If the cardinality of $\mathcal{I}_s$ and $\mathcal{I}_w$ is finite, so that there are finitely many vectors s_n and w_n, then we denote by $\mathcal{S}$ and $\mathcal{W}$ the spaces spanned by the vectors $\{s_n\}$ and $\{w_n\}$, respectively. We refer to $\mathcal{S}$ as the sampling space and to $\mathcal{W}$ as the reconstruction space. If there are infinitely many vectors, then care must be taken to ensure that the sampling is stable so that the sequence of samples obtained by $c[n] = \langle s_n, x \rangle$ has finite energy for any finite-energy $x \in \mathcal{H}$. In order for the reconstruction to be well defined, we must also have that the sum $\sum_n d[n] w_n$ converges. To ensure stability of the sampling and reconstruction, we assume that

[1] An oblique projection is a projection operator E satisfying $E^2 = E$ that is not necessarily Hermitian. The notation $E_{\mathcal{WS}^\perp}$ denotes an oblique projection with range space $\mathcal{W}$ and null space $\mathcal{S}^\perp$. If $\mathcal{W} = \mathcal{S}$, then $E_{\mathcal{WS}^\perp} = P_{\mathcal{W}}$.

the sampling vectors form a *frame* (see Definition 1 below) for their closed span, which we denote by $\mathcal{S}$, and similarly that the reconstruction vectors form a frame for their closed span, which we denote by $\mathcal{W}$.

Definition 1 *([23]).* A family of vectors $\{h_n\}$ in a Hilbert space $\mathcal{H}$ is called a frame for a subspace $\mathcal{A} \subseteq \mathcal{H}$ if there exist constants $0 < A \leq B < \infty$ such that

$$A\|y\|^2 \leq \sum_n |\langle y, h_n \rangle|^2 \leq B\|y\|^2, \tag{5}$$

for all $y \in \mathcal{A}$.

The lower bound in (5) ensures that the vectors $\{h_n\}$ span $\mathcal{A}$; thus the number of frame elements, which we denote by N, must be at least as large as the dimension of $\mathcal{A}$. If $N < \infty$, then the right hand inequality of (5) is always satisfied with $B = \sum_n \langle h_n, h_n \rangle$. Thus, any finite set of vectors that spans $\mathcal{A}$ is a frame for $\mathcal{A}$. If the bounds in (5) satisfy $A = B$, then the frame is called a tight frame.

If the sampling vectors $\{s_n\}$ form a frame for $\mathcal{S}$, then it follows immediately from the upper bound (5) that the sequence of samples $c[n] = \langle s_n, x \rangle$ is in ℓ_2 for any signal x that has bounded norm, and therefore the sampling process is stable. Similarly, if the vectors $\{w_n\}$ form a frame, then the sum $\sum_n d[n] w_n$ converges for any sequence $d \in \ell_2$ [24].

Using set transformations we can express the sequence of samples c as $c = S^* x$, and the reconstruction $\hat{x}$ as

$$\hat{x} = Wd = WHS^* x, \tag{6}$$

where $S : \ell_2 \to \mathcal{H}$ and $W : \ell_2 \to \mathcal{H}$ are the set transformations corresponding to the vectors $\{s_n\}$ and $\{w_n\}$, respectively. The sampling and reconstruction scheme is illustrated in Fig. 1. Note that since $\{s_n\}$

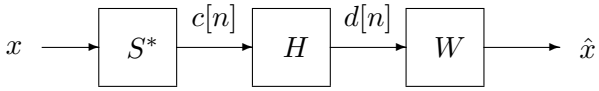


Figure 1: General sampling and reconstruction scheme.

and $\{w_n\}$ are frame sequences, the transformations S and W are bounded. Therefore, if H in Fig. 1 is also chosen to be bounded, then the overall sampling and reconstruction scheme is stable.

If x does not lie in $\mathcal{W}$, then it cannot be perfectly reconstructed using only vectors in $\mathcal{W}$ since $\hat{x}$ given by (4) will always lie in $\mathcal{W}$. Therefore, our problem is to choose the transformation H in Fig. 1 so that $\hat{x}$ is a good approximation of x in the squared-norm sense. If $x \in \mathcal{W}$ and

$$\mathcal{H} = \mathcal{W} \oplus \mathcal{S}^{\perp}, \tag{7}$$

where $\oplus$ denotes the direct sum[2], then x can be perfectly reconstructed from the samples $c[n]$ by choosing H as the bounded operator $H = (S^*W)^\dagger$ [7, 10, 12]; the fact that H is bounded follows from (7) [12]. Indeed, with this choice of H it can be shown that the reconstruction $\hat{x}$ is equal to

$$\hat{x} = E_{\mathcal{W}\mathcal{S}^\perp} x. \tag{8}$$

If $x \notin \mathcal{W}$, then the reconstruction of (8) is no longer equal to x. Nonetheless it has the property that it is a *consistent reconstruction* [1] of x, namely, it yields the same samples as x: $S^*x = S^*\hat{x}$. In fact, under the direct sum condition (7), it is the only reconstruction that is consistent [7, 10]. However, the fact that x and $\hat{x}$ of (8) yield the same samples, does not guarantee that $\hat{x}$ is close to x in the squared-norm sense. Using the fact that $x = E_{\mathcal{W}\mathcal{S}^\perp} x + E_{\mathcal{S}^\perp\mathcal{W}} x$, it follows that the reconstruction error is $\hat{x} - x = E_{\mathcal{S}^\perp\mathcal{W}} x$, which can have arbitrarily large norm if $\mathcal{S}$ is close to $\mathcal{W}^\perp$.

To develop a reconstruction $\hat{x}$ that is close to x, instead of requiring a consistent reconstruction as in [1, 7, 10], we may seek a reconstruction that minimizes the squared-norm of the error. In this case, the transformation H is the solution to the problem

$$\arg\min_H \|\hat{x} - x\|^2 = \arg\min_H \|WHS^*x - x\|^2. \tag{9}$$

To derive the transformation H that is the solution to (9), we first express the objective as

$$\|WHS^*x - x\|^2 = \|WHS^*x - P_{\mathcal{W}}x - P_{\mathcal{W}^\perp}x\|^2 = \|WHS^*x - P_{\mathcal{W}}x\|^2 + \|P_{\mathcal{W}^\perp}x\|^2, \tag{10}$$

where we used the fact that $WHS^*x \in \mathcal{W}$. Thus, our problem becomes

$$\arg\min_H \|(WHS^* - P_{\mathcal{W}})x\|^2 = \arg\min_H \|A(H)x\|^2, \tag{11}$$

where $A(H) \triangleq WHS^* - P_{\mathcal{W}}$.

If $\mathcal{S} = \mathcal{W}$, then we can choose $H = (S^*W)^\dagger$, which results in $WHS^* = E_{\mathcal{W}\mathcal{S}^\perp} = P_{\mathcal{W}}$. Thus, in this case $A(H) = 0$ and the minimal achievable error is $\|P_{\mathcal{W}^\perp}x\|^2$. However, if $\mathcal{W} \neq \mathcal{S}$, then, as we show below, the minimal norm is no longer achievable in general.

To develop a solution to (11) for the case in which $\mathcal{W} \neq \mathcal{S}$, we note that if $x \in \mathcal{S}^\perp$, then $A(H)x = P_{\mathcal{W}}x$ for all choices of H. If, on the other hand, $x \notin \mathcal{S}^\perp$, then we can always choose H such that $A(H)x = 0$. For

[2] $\mathcal{W} \oplus \mathcal{S}^\perp$ is the sum set $\{w + v;\ w \in \mathcal{W}, v \in \mathcal{S}^\perp\}$ with the property $\mathcal{W} \cap \mathcal{S}^\perp = \{0\}$.

example, we may choose

$$H = \frac{1}{\langle S^*x, S^*x\rangle}(W^*W)^\dagger W^*x(S^*x)^*, \tag{12}$$

where for two sequences $a, b \in \ell_2$, the operator $T = ab^*$ is an operator from ℓ_2 to ℓ_2 which is defined by $g = Tr = \langle b, r\rangle a$ so that $g[n] = a[n]\sum_m b^*[m]r[m]$. Indeed, with this choice of H,

$$WHS^*x = \frac{1}{\langle S^*x, S^*x\rangle}P_{\mathcal{W}}x\langle S^*x, S^*x\rangle = P_{\mathcal{W}}x. \tag{13}$$

It follows that H minimizes $\|A(H)x\|$ for all x if and only if

$$A(H)x = \begin{cases} P_{\mathcal{W}}x, & x \in \mathcal{S}^\perp; \\ 0, & x \notin \mathcal{S}^\perp. \end{cases} \tag{14}$$

The minimal square-error then follows from (10) as

$$\min_H \|WHS^*x - x\|^2 = \begin{cases} \|P_{\mathcal{W}^\perp}x\|^2 + \|P_{\mathcal{W}}x\|^2 = \|x\|^2, & x \in \mathcal{S}^\perp; \\ \|P_{\mathcal{W}^\perp}x\|^2, & x \notin \mathcal{S}^\perp. \end{cases} \tag{15}$$

We conclude that any operator H which is a solution to (9) must satisfy (14). As we now show, any operator H satisfying (14) is nonlinear, if $\mathcal{S} \neq \mathcal{W}$.

Indeed, let $x = \alpha x_1 + \beta x_2$ where $x_1 \in \mathcal{S}^\perp$ and $x_1 \notin \mathcal{W}^\perp$ (which is possible if $\mathcal{S} \neq \mathcal{W}$), $x_2 \notin \mathcal{S}^\perp$, and $\alpha, \beta \neq 0$ are chosen such that $x \notin \mathcal{S}^\perp$. Since $x \notin \mathcal{S}^\perp$, we have that

$$WHS^*x = P_{\mathcal{W}}x = \alpha P_{\mathcal{W}}x_1 + \beta P_{\mathcal{W}}x_2. \tag{16}$$

On the other hand,

$$\alpha WHS^*x_1 + \beta WHS^*x_2 = \beta P_{\mathcal{W}}x_2, \tag{17}$$

since $x_1 \in \mathcal{S}^\perp$. If H is linear, then (16) and (17) must be equal, which implies that $P_{\mathcal{W}}x_1 = 0$. However, since $x_1 \notin \mathcal{W}^\perp$, we have $P_{\mathcal{W}}x_1 \neq 0$, so that H is not a linear operator.

We conclude that assuming $\mathcal{S} \neq \mathcal{W}$, an operator H which is the solution to (9) must be a non-linear operator, and leads to an error not smaller than

$$e_{\mathrm{OPT}} \triangleq \|P_{\mathcal{W}^\perp}x\|^2. \tag{18}$$

Thus, if we restrict our attention to linear operators, then we cannot design an operator to minimize the squared-norm error. An exception is the case in which we know that x lies in an a subspace $\mathcal{A}$ of $\mathcal{H}$ such

that $\mathcal{H} = \mathcal{A} \oplus \mathcal{S}^{\perp}$. In this case, we show in Section 4 that the minimal error reconstruction can be obtained for any $x \in \mathcal{A}$ using a linear operator.

If x is not restricted to a subspace, then we can no longer minimize the squared-error directly. We may then consider a different approach where we seek to minimize the worst possible squared-error, over all bounded inputs x, so that H is the solution to

$$\arg\min_{H} \max_{\|x\| \leq L} \|x - \hat{x}\|^2 = \arg\min_{H} \max_{\|x\| \leq L} \|x - WHS^*x\|^2 . \tag{19}$$

Unfortunately, as we show in Section 3, the criterion (19) is overpessimistic and yields the trivial solution $H = 0$. We therefore propose an alternative approach in which instead of minimizing the worst-case squared-norm error, we consider minimizing the worst-case *regret*, which is the difference between the squared-norm error and the smallest possible squared-norm error achievable with the non-linear H, namely e_{OPT} of (18). By considering the difference between the error and the best possible error, rather than the error itself, we may counterbalance the conservative behavior of the minimax approach. Thus, we seek the reconstruction $\hat{x} = WHS^*x$ that is the solution to

$$\min_{\hat{x}} \max_{\|x\| \leq L} \left\{ \|x - \hat{x}\|^2 - \|P_{\mathcal{W}^{\perp}} x\|^2 \right\} . \tag{20}$$

The minimax regret reconstruction is developed in Section 3.2. As we show, the resulting reconstruction is given by $\hat{x} = P_{\mathcal{W}} P_{\mathcal{S}} x$.

3 Minimax Reconstruction

3.1 Minimax Squared-Error

We begin by showing that the minimax transformation H that is the solution to

$$\arg\min_{H} \max_{\|x\| \leq L} \|x - WHS^*x\|^2 \tag{21}$$

is given by $H = 0$. To this end, we first develop a lower bound on the smallest value of $\max_{\|x\| \leq L} \|x - WHS^*x\|^2$, and then show that this bound is achievable with $H = 0$.

Specifically,

$$\max_{\|x\| \leq L} \|\hat{x} - x\|^2 \geq \max_{\|x\| \leq L, x \in \mathcal{S}^{\perp}} \|\hat{x} - x\|^2 . \tag{22}$$

Since for $x \in \mathcal{S}^{\perp}$ we have that $\hat{x} = 0$,

$$\min_{H} \max_{\|x\| \leq L} \|x - \hat{x}\|^2 \geq \max_{\|x\| \leq L, x \in \mathcal{S}^{\perp}} \|x\|^2 = L^2. \tag{23}$$

We conclude that for any choice of H, $\max_{\|x\| \leq L} \|x - WHS^*x\|^2 \geq L^2$. For $H = 0$,

$$\max_{\|x\| \leq L} \|x - WHS^*x\|^2 = \max_{\|x\| \leq L} \|x\|^2 = L^2, \tag{24}$$

and therefore $H = 0$ is the solution to (21).

Since the minimax criterion leads to the trivial solution $H = 0$, we consider, in the next section, a competitive approach in which instead of minimizing the worst-case squared-norm error, we seek to minimize the worst-case regret.

3.2 Minimax Regret Reconstruction

We now consider minimizing the worst-case regret, namely the worst-case difference between the squared-norm error and the smallest attainable error $e_{\text{OPT}} = \|P_{\mathcal{W}^{\perp}} x\|^2$. Thus, we seek H that is the solution to the problem

$$\arg\min_{H} \max_{\|x\| \leq L} \left\{ \|x - \hat{x}\|^2 - \|P_{\mathcal{W}^{\perp}} x\|^2 \right\}. \tag{25}$$

Expressing x as $x = P_{\mathcal{W}} x + P_{\mathcal{W}^{\perp}} x$, we have that

$$\|x - \hat{x}\|^2 = \|P_{\mathcal{W}} x - \hat{x} + P_{\mathcal{W}^{\perp}} x\|^2 = \|P_{\mathcal{W}} x - \hat{x}\|^2 + \|P_{\mathcal{W}^{\perp}} x\|^2, \tag{26}$$

where we used the fact that $\hat{x} \in \mathcal{W}$. Therefore, the problem (25) becomes

$$\arg\min_{H} \max_{\|x\| \leq L} \|P_{\mathcal{W}} x - \hat{x}\|^2. \tag{27}$$

The solution to (27) is given in the following theorem.

Theorem 1. *Let H_{REG} be the solution to*

$$\arg\min_{H} \max_{\|x\| \leq L} \left\{ \|x - \hat{x}\|^2 - \|P_{\mathcal{W}^{\perp}} x\|^2 \right\} = \arg\min_{H} \max_{\|x\| \leq L} \|WHS^*x - P_{\mathcal{W}} x\|^2.$$

Then H_{REG} is independent of L and is given by

$$H_{\text{REG}} = (W^*W)^{\dagger} W^* S (S^*S)^{\dagger}.$$

The resulting reconstruction is $\hat{x} = P_{\mathcal{W}} P_{\mathcal{S}} x$.

We note that if $\mathcal{W} \subseteq \mathcal{S}$, or $x \in \mathcal{S}$, then $\hat{x} = P_{\mathcal{W}} P_{\mathcal{S}} x = P_{\mathcal{W}} x$ which is the minimal-error approximation of x in $\mathcal{W}$.

Proof. The maximum squared-error norm can be bounded as

$$\max_{\|x\| \leq L} \|\hat{x} - P_{\mathcal{W}} x\|^2 \geq \max_{\|x\| \leq L, x \in \mathcal{S}^\perp} \|\hat{x} - P_{\mathcal{W}} x\|^2 . \tag{28}$$

Since for $x \in \mathcal{S}^\perp$ we have that $\hat{x} = 0$,

$$\min_H \max_{\|x\| \leq L} \|P_{\mathcal{W}} x - \hat{x}\|^2 \geq \max_{\|x\| \leq L, x \in \mathcal{S}^\perp} \|P_{\mathcal{W}} x\|^2 = \max_{\|x\| \leq L} \|P_{\mathcal{W}} P_{\mathcal{S}^\perp} x\|^2 . \tag{29}$$

We now show that with $H = H_{\text{REG}}$, the lower bound in (29) is achieved. Indeed, if $H = H_{\text{REG}}$, then

$$\hat{x} = WHS^* = W \left(W^* W\right)^\dagger W^* S \left(S^* S\right)^\dagger S^* = P_{\mathcal{W}} P_{\mathcal{S}}, \tag{30}$$

and

$$\|P_{\mathcal{W}} x - \hat{x}\|^2 = \|P_{\mathcal{W}} (I - P_{\mathcal{S}}) x\|^2 = \|P_{\mathcal{W}} P_{\mathcal{S}^\perp} x\|^2 . \tag{31}$$

□

Using results from [12] we can show that the minimax regret transformation H_{REG} of Theorem 1 is bounded. In particular, we have the following lemma.

Lemma 1. *[12] Let* $S : \ell_2 \to \mathcal{H}$ *and* $W : \ell_2 \to \mathcal{H}$ *be bounded transformations on* $\mathcal{H}$ *with* $\mathcal{R}(S) = \mathcal{S}$ *and* $\mathcal{R}(W) = \mathcal{W}$*, where* $\mathcal{H} = \mathcal{W} \oplus \mathcal{S}^\perp$*. Then* $(S^* W)^\dagger$ *is a bounded operator from* ℓ_2 *to* ℓ_2*.*

From Lemma 1 it follows that both of the operators $(W^* W)^\dagger$ and $(S^* S)^\dagger$ are bounded, rendering H_{REG} a bounded operator.

An interesting feature of the minimax regret reconstruction $\hat{x}$ of Theorem 1 is that it does not depend on the norm bound L. Therefore, $\hat{x} = P_{\mathcal{W}} P_{\mathcal{S}} x$ minimizes the worst-case regret error over all bounded inputs x, regardless of the norm of x. Furthermore, in the derivation of the minimax regret reconstruction we did not impose any constraints on the spaces $\mathcal{W}$ and $\mathcal{S}$. In contrast, the consistent reconstruction of [7, 10, 12] assumes explicitly that $\mathcal{W}$ and $\mathcal{S}$ satisfy $\mathcal{H} = \mathcal{W} \oplus \mathcal{S}^\perp$.

Another desirable property of the minimax regret reconstruction is that the resulting reconstruction error is always bounded by twice the norm of x. Specifically, expressing the reconstruction error as

$$\hat{x} - x = P_{\mathcal{W}} P_{\mathcal{S}} x - P_{\mathcal{W}} x - P_{\mathcal{W}^\perp} x = -P_{\mathcal{W}} P_{\mathcal{S}^\perp} x - P_{\mathcal{W}^\perp} x, \tag{32}$$

we have that

$$\|\hat{x} - x\|^2 = \|P_{\mathcal{W}} P_{\mathcal{S}^\perp} x\|^2 + \|P_{\mathcal{W}^\perp} x\|^2 \leq 2\|x\|^2. \tag{33}$$

We note that tighter bounds on the reconstruction error are derived in Section 6.

3.3 Geometric Illustration of the Reconstruction

We now illustrate the minimax regret reconstruction geometrically. To this end, we first note that sampling x with sampling vectors in $\mathcal{S}$, is equivalent to sampling the orthogonal projection of x onto $\mathcal{S}$, denoted by $x_{\mathcal{S}} = P_{\mathcal{S}} x$. This follows from the relation

$$\langle s_n, x \rangle = \langle P_{\mathcal{S}} s_n, x \rangle = \langle s_n, P_{\mathcal{S}} x \rangle. \tag{34}$$

Since $x_{\mathcal{S}} \in \mathcal{S}$ and the vectors $\{s_n\}$ span $\mathcal{S}$, $x_{\mathcal{S}}$ is uniquely determined by the samples $c[n]$. Therefore, knowing $c[n]$ is equivalent to knowing $x_{\mathcal{S}}$. The reconstruction problem then becomes the problem of reconstructing an arbitrary signal in $\mathcal{H}$ from its orthogonal projection $x_{\mathcal{S}}$ onto a subspace $\mathcal{S}$ of $\mathcal{H}$, where the reconstruction is constrained to lie in a subspace $\mathcal{W}$ of $\mathcal{H}$. Fig. 2(a) depicts the orthogonal projection of an unknown signal $x \in \mathcal{H}$ onto $\mathcal{S}$, denoted $x_{\mathcal{S}}$. The minimax regret reconstruction chooses as the reconstructed signal the orthogonal projection of $x_{\mathcal{S}}$ onto $\mathcal{W}$, as illustrated in Fig. 2(b).

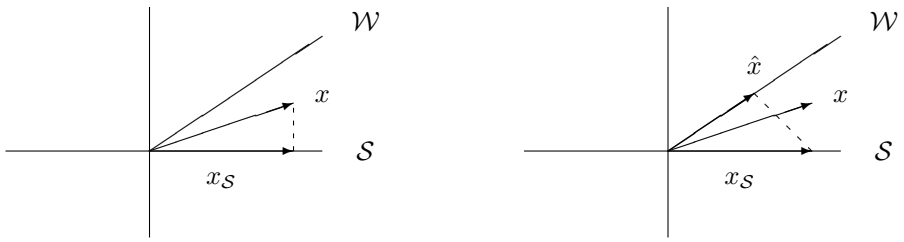


Figure 2: Illustration of minimax regret reconstruction of x from $x_{\mathcal{S}} = P_{\mathcal{S}} x$ (a) orthogonal projection of x onto $\mathcal{S}$ (b) minimax regret reconstruction $\hat{x} = P_{\mathcal{W}} P_{\mathcal{S}} x$.

4 Minimal Error Reconstruction on a Subspace

We have seen in Section 2, that if $\mathcal{S} \neq \mathcal{W}$, then the norm $\|x - \hat{x}\|$ can be minimized and the minimal error can be achieved *for all* x only if a non-linear transformation H is applied. Therefore, to develop a linear reconstruction transformation, we considered a minimax regret criterion. We also showed that if we

know that $x \in \mathcal{W}$ and $\mathcal{H} = \mathcal{W} \oplus \mathcal{S}^\perp$, then the minimal error can be achieved, since x can be perfectly reconstructed. In this section, we extend this result to a more general class of subspaces. Specifically, we consider the case in which x is known to lie in a subspace $\mathcal{A}$ such that $\mathcal{H} = \mathcal{A} \oplus \mathcal{S}^\perp$, and show that in this case, the minimal error reconstruction can be achieved.

Thus, we consider the problem

$$\min_H \|WHS^*x - x\|^2, \quad x \in \mathcal{A}. \tag{35}$$

Since $\mathcal{A} \cap \mathcal{S}^\perp = \{0\}$, it follows from the discussion in Section 2 that the smallest error $e_{\text{OPT}} = \|P_{\mathcal{W}^\perp}x\|^2$ is achievable using an input dependent H as in (12). Theorem 2 below establishes that there also exists a *linear* H that achieves the minimal norm reconstruction.

Theorem 2. *Let $H_{\mathcal{A}}$ be a solution to*

$$\arg\min_H \|x - WHS^*x\|^2, \quad x \in \mathcal{A}$$

where $\mathcal{A}$ is an arbitrary subspace such that $\mathcal{H} = \mathcal{A} \oplus \mathcal{S}^\perp$. Then

$$H_{\mathcal{A}} = (W^*W)^\dagger W^*A\,(S^*A)^\dagger, \tag{36}$$

where A is a bounded operator with $\mathcal{R}(A) = \mathcal{A}$. The resulting reconstruction $\hat{x}$ is the minimal-norm reconstruction $\hat{x} = P_{\mathcal{W}}x$.

Before proving the theorem, we note that from Lemma 1 the pseudoinverse $(S^*A)^\dagger$ is a well defined bounded operator [12]. Furthermore, the operator $A(S^*A)^\dagger$ in (36) is independent of the choice of A, as incorporated in the following lemma [22].

Lemma 2. *Let $\mathcal{A}$ and $\mathcal{S}$ be closed subspaces of a Hilbert space $\mathcal{H}$ such that $\mathcal{H} = \mathcal{A} \oplus \mathcal{S}^\perp$, and let S be a bounded operator with $\mathcal{R}(S) = \mathcal{S}$. Then the operator*

$$M = A(S^*A)^\dagger$$

is independent of the choice of the bounded operator $A : \ell^2 \to \mathcal{H}$, as long as $\mathcal{R}(A) = \mathcal{A}$.

We now prove Theorem 2.

Proof. With the suggested $H = H_{\mathcal{A}}$, the reconstruction is

$$\hat{x} = WHS^*x = P_{\mathcal{W}}A\,(S^*A)^\dagger\,S^*x = P_{\mathcal{W}}E_{\mathcal{A}\mathcal{S}^\perp}x = P_{\mathcal{W}}x = P_{\mathcal{W}}x, \tag{37}$$

where we used the fact that from (2), $A\,(S^*A)^\dagger\,S^* = E_{\mathcal{A}\mathcal{S}^\perp}$, and since $x \in \mathcal{A}$, $E_{\mathcal{A}\mathcal{S}^\perp}x = x$. □

We now consider several special choices of $\mathcal{A}$. First, suppose that $\mathcal{A} = \mathcal{S}$. In this case, the condition $\mathcal{H} = \mathcal{A} \oplus \mathcal{S}^\perp$ is always satisfied. From Theorem 2,

$$H_{\mathcal{A}} = (W^*W)^\dagger W^* S (S^*S)^\dagger, \tag{38}$$

which is equivalent to the minimax regret transformation of Theorem 1. Thus, we conclude that the minimax regret transformation minimizes the squared-norm error over all values of $x \in \mathcal{S}$. Indeed, $\hat{x} = W H_{\text{REG}} S^* x = P_{\mathcal{W}} P_{\mathcal{S}} x = P_{\mathcal{W}} x$ for all $x \in \mathcal{S}$.

As another example, we can choose $\mathcal{A} = \mathcal{W}$, as long as $\mathcal{H} = \mathcal{W} \oplus \mathcal{S}^\perp$. Then

$$H_{\mathcal{A}} = (W^*W)^\dagger W^* W (S^*W)^\dagger = P_{\mathcal{N}(\mathcal{W})^\perp} (S^*W)^\dagger = (S^*W)^\dagger, \tag{39}$$

where we used the fact that $\mathcal{R}\left((S^*W)^\dagger\right) = \mathcal{N}(S^*W)^\perp = \mathcal{N}(W)^\perp$. Thus, in this case $H_{\mathcal{A}}$ is equal to the consistent reconstruction transformation, which agrees with the fact that the consistent reconstruction minimizes the squared-norm error over all values of $x \in \mathcal{W}$.

In Theorem 2 we assumed that $\mathcal{H} = \mathcal{A} \oplus \mathcal{S}^\perp$, which implies that $\mathcal{A} \cap \mathcal{S}^\perp = \{0\}$. If $\mathcal{A} \cap \mathcal{S}^\perp \neq \{0\}$, then following the same steps as in Section 2 we can show that the transformation H minimizing the squared-norm error must be nonlinear. Furthermore, since there exists an $x \in \mathcal{A} \cap \mathcal{S}^\perp$ such that $x \neq 0$, the derivation in Sections 3.1 and 3.2 can be used to show that the minimax squared-error reconstruction over $x \in \mathcal{A}$ is $H = 0$, and the minimax regret reconstruction over $x \in \mathcal{A}$ is the same as that given in Theorem 1.

4.1 Geometric Interpretation

We have seen in this section that if we know that x is in $\mathcal{A}$, where $\mathcal{A} \oplus \mathcal{S}^\perp = \mathcal{H}$, then we can always obtain the minimal norm reconstruction $\hat{x} = P_{\mathcal{W}} x$ given the samples $c[n]$. This result can be understood by considering a geometric interpretation of the sampling and reconstruction. Specifically, in Section 3.3 we showed that knowledge of the signal samples $c[n]$ is equivalent to knowledge of the orthogonal projection of the signal onto the subspace $\mathcal{S}$. In Fig. 3 we illustrate the fact that there is only one vector in $\mathcal{A}$ whose orthogonal projection onto $\mathcal{S}$ is $x_{\mathcal{S}}$. Thus, given $x_{\mathcal{S}}$, we can immediately determine the original vector x, if we know that x is in $\mathcal{A}$. In our setup we are constrained to obtain a reconstruction in $\mathcal{W}$. But, since we can determine x from $x_{\mathcal{S}}$ in principle, we can also determine its orthogonal projection onto $\mathcal{W}$, which is the minimal norm reconstruction.

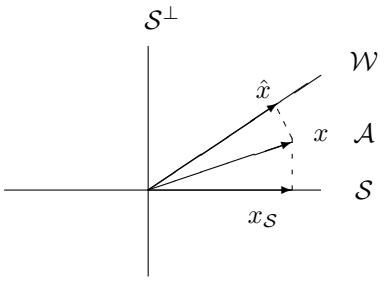


Figure 3: Illustration of minimal-norm reconstruction $\hat{x} = P_\mathcal{W} x$ of $x \in \mathcal{A}$ from $x_\mathcal{S} = P_\mathcal{S} x$, with $\mathcal{H} = \mathcal{A} \oplus \mathcal{S}^\perp$.

5 Reconstruction in Shift-Invariant Spaces

The reconstruction algorithms of Theorems 1 and 2 were derived for arbitrary subspaces of an arbitrary Hilbert space. As we now show, in the special case in which $\mathcal{H} = L_2$ and $\mathcal{S}$ and $\mathcal{W}$ are real SI subspaces, $H_{\rm REG}$ and $H_\mathcal{A}$ reduce to discrete-time LTI filters, and can therefore be implemented efficiently. Furthermore, in this case the sampling and reconstruction can be implemented using continuous-time LTI filters.

Suppose that $x = x(t)$ is a real signal in L_2, and that

$$\begin{aligned} \mathcal{S} &= \left\{ f(t) | f(t) = \sum_n a[n] s(t-n), a \in \ell_2 \right\}; \\ \mathcal{W} &= \left\{ f(t) | f(t) = \sum_n a[n] w(t-n), a \in \ell_2 \right\}, \end{aligned} \tag{40}$$

where $s(t)$ and $w(t)$ are the generators of $\mathcal{S}$ and $\mathcal{W}$, respectively. To ensure that the vectors $\{s_n(t) = s(t-n)\}$ and $\{w_n(t) = w(t-n)\}$ form frames for $\mathcal{S}$ and $\mathcal{W}$ respectively, we must have that [25]

$$\begin{aligned} \alpha \le \textstyle\sum_n |W(\omega + 2\pi k)|^2 \le \beta, \quad \omega \in \mathcal{I}_\mathcal{W}; \\ \gamma \le \textstyle\sum_n |S(\omega + 2\pi k)|^2 \le \eta, \quad \omega \in \mathcal{I}_\mathcal{S}, \end{aligned} \tag{41}$$

for some $\alpha > 0$ and $\gamma > 0$, where $W(\omega)$ and $S(\omega)$ are the continuous-time Fourier transforms of the generators $s(t)$ and $w(t)$ respectively and $\mathcal{I}_\mathcal{W}, \mathcal{I}_\mathcal{S}$ are the set of frequencies ω for which $\sum_k |W(\omega + 2\pi k)|^2 \neq 0$ and $\sum_k |S(\omega + 2\pi k)|^2 \neq 0$, respectively.

In this case, the samples $c[n]$, which are given by

$$c[n] = \int s(t-n) x(t) dt = x(t) * s(-t)|_{t=n}, \tag{42}$$

correspond to samples at times $t = n$ of the output of a filter with impulse response $s(-t)$ with $x(t)$ as its input. Here $g(t) * z(t)$ denotes the convolution between the signals $g(t)$ and $z(t)$, and $y(t)|_{t=n} \triangleq y(t = n)$.

The reconstructed signal corresponds to the output of a filter with impulse response $w(t)$, with an impulse train whose values are the corrected measurements $d[n]$ as its input where, $d = H_{\text{REG}}c$ or $d = H_{\mathcal{A}}c$. We now show that in both cases the corrected samples $d[n]$ can be obtained by filtering the samples $c[n]$ with a discrete-time LTI filter. To this end we rely on the following lemma.

Lemma 3. *Let G and Z denote set transformations corresponding to frames $\{g_n(t) = g(t-n)\}$ and $\{z_n(t) = z(t-n)\}$. Then G^*Za is equivalent to filtering the sequence $a \in \ell_2$ with a discrete-time LTI filter with frequency response*

$$H(\omega) = \sum_{k=-\infty}^{\infty} G^*(\omega + 2\pi k)Z(\omega + 2\pi k), \tag{43}$$

where $G(\omega)$ and $Z(\omega)$ denote the continuous-time Fourier transforms of $g(t)$ and $z(t)$, respectively.

Proof. Denoting $b = G^*Za$, we have that

$$b[m] = \langle g(t-m), Za\rangle = \langle g(t-m), \sum_n a[n]z(t-n)\rangle = \sum_n a[n]\langle g(t-m), z(t-n)\rangle. \tag{44}$$

Now,

$$\langle g(t-m), z(t-n)\rangle = \int_{-\infty}^{\infty} g(t-m)z(t-n)dt = \int_{-\infty}^{\infty} g(t)z(t+m-n)dt \stackrel{\triangle}{=} h[n-m], \tag{45}$$

where

$$h[n] = \int_{-\infty}^{\infty} g(t)z(t-n)dt = g(t) * z(-t)|_{t=n}. \tag{46}$$

Substituting (45) into (44) we have that

$$b[m] = \sum_n a[n]h[n-m] = a[m] * h[-m], \tag{47}$$

where $a[m] * h[-m]$ is the discrete-time convolution between the sequences $a[m]$ and $h[-m]$. Thus, the sequence $b[n]$ defined by $b = G^*Za$ is equal to the convolution of $a[n]$ with a filter with impulse response $h[-n]$ given by (46). Applying the Poisson sum formula [26] to (46) results in the frequency response given by (43). □

We first consider the case in which $d = H_{\text{REG}}c = (W^*W)^\dagger W^*S(S^*S)^\dagger c$, where S and W are the set transformations corresponding to $\{s_n(t) = s(t-n)\}$ and $\{w_n(t) = w(t-n)\}$, respectively. From Lemma 3 it follows that W^*S is equivalent to a filter with frequency response

$$Q(\omega) = \sum_{k=-\infty}^{\infty} S(\omega + 2\pi k)W^*(\omega + 2\pi k). \tag{48}$$

Similarly, S^*S is equivalent to a filter with frequency response $\sum_{k=-\infty}^{\infty} |S(\omega + 2\pi k)|^2$. It then follows

immediately that $(S^*S)^\dagger$ is equivalent to a filter with frequency response

$$T_{\mathcal{S}}(\omega) = \begin{cases} \frac{1}{\sum_{k=-\infty}^{\infty} |S(\omega+2\pi k)|^2}, & \omega \in \mathcal{I}_{\mathcal{S}}; \\ 0, & \omega \notin \mathcal{I}_{\mathcal{S}}. \end{cases} \tag{49}$$

Likewise, $(W^*W)^\dagger$ is equivalent to a filter with frequency response

$$T_{\mathcal{W}}(\omega) = \begin{cases} \frac{1}{\sum_{k=-\infty}^{\infty} |W(\omega+2\pi k)|^2}, & \omega \in \mathcal{I}_{\mathcal{W}}; \\ 0, & \omega \notin \mathcal{I}_{\mathcal{W}}. \end{cases} \tag{50}$$

Combining (48), (49) and (50), it follows that the corrected samples $d = H_{\mathrm{REG}}c$ can be obtained from c using a discrete-time filter with frequency response

$$H(\omega) = T_{\mathcal{S}}(\omega)Q(\omega)T_{\mathcal{W}}(\omega) = \begin{cases} \frac{\sum_{k=-\infty}^{\infty} S(\omega+2\pi k)W^*(\omega+2\pi k)}{\sum_{k=-\infty}^{\infty} |S(\omega+2\pi k)|^2 \sum_{k=-\infty}^{\infty} |W(\omega+2\pi k)|^2}, & \omega \in \mathcal{I}_{\mathcal{W}} \cap \mathcal{I}_{\mathcal{S}}; \\ 0, & \omega \notin \mathcal{I}_{\mathcal{W}} \cap \mathcal{I}_{\mathcal{S}}. \end{cases} \tag{51}$$

The sampling scheme of Fig. 1 then reduces to the scheme depicted in Fig. 4.

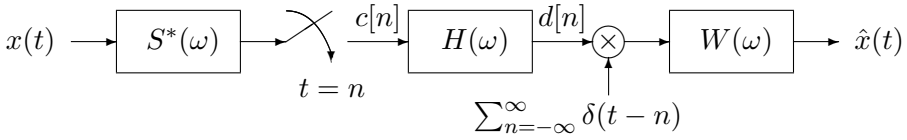


Figure 4: Minimax regret reconstruction of $x(t)$ in SI spaces. The frequency response of the filter $H(\omega)$ is given by (51).

Note that the regret reconstruction filter (51) has the property that its magnitude is no larger than 1. Indeed, from the Cauchy-Schwartz inequality we have that

$$\left| \sum_{k=-\infty}^{\infty} S(\omega+2\pi k)W^*(\omega+2\pi k) \right| \leq \sum_{k=-\infty}^{\infty} |S(\omega+2\pi k)|^2 \sum_{k=-\infty}^{\infty} |W(\omega+2\pi k)|^2 , \tag{52}$$

so that $|H(\omega)| \leq 1$.

The consistent reconstruction scheme for SI spaces, developed in [1], has the same form as Fig. 4, where the filter $H(\omega)$ is specified by

$$H(\omega) = \begin{cases} \frac{1}{\sum_{k=-\infty}^{\infty} S^*(\omega+2\pi k)W(\omega+2\pi k)}, & \omega \in \mathcal{I}_{\mathcal{W}} \cap \mathcal{I}_{\mathcal{S}}; \\ 0, & \omega \notin \mathcal{I}_{\mathcal{W}} \cap \mathcal{I}_{\mathcal{S}}. \end{cases} \tag{53}$$

If $\mathcal{H} = \mathcal{W} \oplus \mathcal{S}^\perp$, then $\mathcal{I}_{\mathcal{W}} = \mathcal{I}_{\mathcal{S}}$ and $\left|\sum_{k=-\infty}^{\infty} S^*(\omega+2\pi k)W(\omega+2\pi k)\right| \geq A > 0$ on $\omega \in \mathcal{I}_{\mathcal{W}}$, so that the filter

is well defined [Proposition 4.8][27]. Note, however, that although the magnitude of the filter is bounded, it can be arbitrarily large. This is in contrast with the magnitude of the regret reconstruction filter, which, as we have seen, is always no larger than 1.

We next consider the case of subspace reconstruction with $H_{\mathcal{A}}$ of Theorem 2, where we assume that $x(t)$ is in a SI subspace $\mathcal{A}$ generated by the function $a(t)$, such that $\mathcal{A} \oplus \mathcal{S}^{\perp} = \mathcal{H}$. Using Lemma 3 it follows that if $d = H_{\mathcal{A}} c$, then $d[n]$ can be obtained by filtering the sequence $c[n]$ by an LTI filter with frequency response

$$H(\omega) = \begin{cases} \frac{\sum_{k=-\infty}^{\infty} A(\omega+2\pi k) W^*(\omega+2\pi k)}{\sum_{k=-\infty}^{\infty} |W(\omega+2\pi k)|^2 \sum_{k=-\infty}^{\infty} A(\omega+2\pi k) S^*(\omega+2\pi k)}, & \omega \in \mathcal{I}_{\mathcal{W}} \cap \mathcal{I}_{\mathcal{S}}; \\ 0, & \omega \notin \mathcal{I}_{\mathcal{W}} \cap \mathcal{I}_{\mathcal{S}}, \end{cases} \tag{54}$$

where $A(\omega)$ is the continuous-time Fourier transform of $a(t)$ and we have used the fact that $\sum_{k=-\infty}^{\infty} A(\omega + 2\pi k) S^*(\omega + 2\pi k) \neq 0$ on $\mathcal{I}_{\mathcal{S}}$, since $\mathcal{A} \oplus \mathcal{S}^{\perp} = \mathcal{H}$.

The filters of (51), (53) and (54) are particularly efficient to implement in the case of spline spaces, which we discuss in the next section.

5.1 Efficient Implementation with Splines

A special class of SI spaces, which is popular in applications, is the class of spline spaces, *i.e.,* SI spaces in which the generating function is a spline [15, 16, 4]. In this section, we consider the case in which the sampling space $\mathcal{S}$ and the reconstruction space $\mathcal{W}$ are spline subspaces, and show that, using the results of [15, 16], the reconstruction filters take on a particularly simple form.

Assume that the sampling space $\mathcal{S}$ and the reconstruction space $\mathcal{W}$ are SI spaces, generated by splines of order n_s and n_w, respectively. Thus

$$\begin{aligned} \mathcal{S} &= \left\{ f(t) \mid f(t) = \sum_n a[n] \beta^{n_s}(t-n), a \in \ell_2 \right\}; \\ \mathcal{W} &= \left\{ f(t) \mid f(t) = \sum_n a[n] \beta^{n_w}(t-n), a \in \ell_2 \right\}, \end{aligned} \tag{55}$$

where $\beta^m(t)$ is a symmetrical B-spline of order m which can be defined recursively by $\beta^{m+1}(t) = \beta^m(t) * \beta^0(t)$ with

$$\beta^0(t) = \begin{cases} 1, & t \in [-0.5, 0.5) \\ 0, & \text{otherwise.} \end{cases} \tag{56}$$

One of the main advantages of using splines, is the ability to compute many operations using the discrete-representatives of the spline functions (see for example [17]). The discrete representative of a continuous spline function $\beta^m(t)$ is the discrete and symmetric *finite impulse response* (FIR) spline filter $b^m[n]$, which is obtained by sampling the continuous spline: $b^m[n] \triangleq \beta^m(t = n)$.

Reconstruction	Filter Impulse Response
Consistent	$\left\{b^{n_s+n_w+1}\right\}^{-1}$
Regret	$\left\{b^{2n_w+1}\right\}^{-1} * b^{n_w+n_s+1} * \left\{b^{2n_s+1}\right\}^{-1}$
Subspace	$\left\{b^{2n_w+1}\right\}^{-1} * b^{n_w+n_a+1} * \left\{b^{n_a+n_s+1}\right\}^{-1}$

Table 1: Implementation of the consistent (53), regret (51) and subspace (54) reconstruction filters for B-spline spaces.

Implementing the filters of (51), (53) and (54) requires computing summations of the form $D(\omega) = \sum_{k=-\infty}^{\infty} S^*(\omega + 2\pi k)W(\omega + 2\pi k)$. This becomes particularly simple in the case of splines. Specifically, the function $D(\omega)$ is the discrete-time Fourier transform of the sequence

$$d[n] = w(t) * s(-t)|_{t=n} = \int_{-\infty}^{\infty} w(t)s(n-t)dt. \tag{57}$$

If $w(t) = \beta^{n_w}(t)$ and $s(t) = \beta^{n_s}(t)$ are splines, then

$$\int_{-\infty}^{\infty} \beta^{n_w}(t)\beta^{n_s}(n-t)dt = \beta^{n_w+n_s+1}(-n) = b^{n_1+n_2+1}[n], \tag{58}$$

where we used the symmetry of the spline function and the convolution property $\beta^{n_1}(t) * \beta^{n_2}(t) = \beta^{n_1+n_2+1}(t)$. Therefore, $D(\omega)$ is simply the transform of $b^{n_1+n_2+1}[n]$, and $1/D(\omega)$ is the transform of $\left\{b^{n_1+n_2+1}\right\}^{-1}[n]$, where $\{b^m\}^{-1}[n]$ denotes the convolutional inverse of the filter $b^m[n]$ *i.e.,* $\{b^m\}^{-1}[n] * b^m[n] = \delta[n]$. Each of the filters (51), (53) and (54) can then be implemented using discrete-time spline filters of appropriate order, as summarized in Table 1. In the table, n_w, n_s and n_a denote the orders of the spline generators of the subspaces $\mathcal{W}$, $\mathcal{S}$ and $\mathcal{A}$, respectively.

Note that when $n_s = n_w$ (*i.e.,* $\mathcal{S} = \mathcal{W}$) all the filters reduce to $\left\{b^{2n_w+1}\right\}^{-1}$ resulting in $\hat{x} = P_{\mathcal{W}}x$.

6 Error Analysis

We now analyze the error $e_{\text{REG}} \triangleq x - P_{\mathcal{W}}P_{\mathcal{S}}x$ of the minimax regret reconstruction, and compare it with the error of the consistent reconstruction $e_{\text{CON}} \triangleq x - E_{\mathcal{W}\mathcal{S}^\perp}x$. In particular, we develop bounds on the norm of the reconstruction error in both cases, and show, that if the spaces $\mathcal{S}$ and $\mathcal{W}$ are far enough apart, or if x has enough energy in $\mathcal{S}$, then the worst-case and best-case error using the minimax regret reconstruction are smaller than the worst-case and best-case error using the consistent reconstruction.

Before proceeding to the detailed development, we note that if $x \in \mathcal{A}$ where $\mathcal{H} = \mathcal{A} \oplus \mathcal{S}^\perp$, then the subspace approach of Theorem 2 will yield the minimal error approximation to x. In particular, if $\mathcal{A} = \mathcal{W}$, then the consistent reconstruction approach will yield perfect reconstruction of x, which is not true for the minimax regret reconstruction. Therefore, if we know that $x \in \mathcal{A}$, then we should use the

subspace reconstruction algorithm. The interesting question is how to reconstruct the signal when x does not necessarily lie in a subspace $\mathcal{A}$ satisfying $\mathcal{H} = \mathcal{A} \oplus \mathcal{S}^{\perp}$. In this case we now show that we can often reduce the error by using the minimax regret reconstruction. These analytical results are also demonstrated through simulation in Section 7.

Since the samples $c = S^*x$ are obtained with sampling vectors that are in $\mathcal{S}$, $c = 0$ for any $x \in \mathcal{S}^{\perp}$, and therefore the reconstruction error for any such x is equal to the norm of x, regardless of the choice of H. Thus, instead of developing bounds on the norm of the reconstruction error, we consider the ratios $\gamma_{\text{REG}} \triangleq \|e_{\text{REG}}\|^2 / \|P_{\mathcal{S}}^{\perp} x\|^2$ and $\gamma_{\text{CON}} \triangleq \|e_{\text{CON}}\|^2 / \|P_{\mathcal{S}}^{\perp} x\|^2$ where we assume explicitly throughout this section that $x \notin \mathcal{S}^{\perp}$.

Expressing e_{REG} as $e_{\text{REG}} = P_{\mathcal{W}^{\perp}} e_{\text{REG}} + P_{\mathcal{W}} e_{\text{REG}}$, we have that

$$\|e_{\text{REG}}\|^2 = \|e_{\text{OPT}}\|^2 + \|P_{\mathcal{W}} P_{\mathcal{S}^{\perp}} x\|^2 , \tag{59}$$

where $e_{\text{OPT}} = \|P_{\mathcal{W}^{\perp}} x\|^2$ is the optimal squared-error. Note that for $x \in \mathcal{S}$, $\|e_{\text{REG}}\|^2 = \|e_{\text{OPT}}\|^2$, so that the minimax regret reconstruction is optimal. If $x \notin \mathcal{S}$, then

$$\gamma_{\text{REG}} = \gamma_{\text{OPT}} + \frac{\|P_{\mathcal{W}} P_{\mathcal{S}^{\perp}} x\|^2}{\|P_{\mathcal{S}^{\perp}} x\|^2}, \tag{60}$$

where we denoted $\gamma_{\text{OPT}} = \|e_{\text{OPT}}\|^2 / \|P_{\mathcal{S}}^{\perp} x\|^2$. Using the fact that $P_{\mathcal{S}^{\perp}} x / \|P_{\mathcal{S}^{\perp}} x\| \in C_{\mathcal{S}^{\perp}}$, where for an arbitrary subspace $\mathcal{A}$, $C_{\mathcal{A}} \triangleq \{a : \ a \in \mathcal{A}, \|a\| = 1\}$ is the union sphere of $\mathcal{A}$, we have that

$$\inf_{v \in C_{\mathcal{S}^{\perp}}} \|P_{\mathcal{W}} v\|^2 \leq \frac{\|P_{\mathcal{W}} P_{\mathcal{S}^{\perp}} x\|^2}{\|P_{\mathcal{S}^{\perp}} x\|^2} \leq \sup_{v \in C_{\mathcal{S}^{\perp}}} \|P_{\mathcal{W}} v\|^2 . \tag{61}$$

Now, from the definition of the angle between two closed subspaces of a Hilbert space[3] [1, 28],

$$\begin{aligned} \inf_{v \in C_{\mathcal{S}^{\perp}}} \|P_{\mathcal{W}} v\|^2 &\triangleq \cos^2\left(\mathcal{S}^{\perp}, \mathcal{W}\right) \\ \sup_{v \in C_{\mathcal{S}^{\perp}}} \|P_{\mathcal{W}} v\|^2 &\triangleq M^2\left(\mathcal{S}^{\perp}, \mathcal{W}\right) = 1 - \cos^2\left(\mathcal{W}, \mathcal{S}\right). \end{aligned} \tag{62}$$

Combining (60), (61) and (62),

$$\cos^2\left(\mathcal{S}^{\perp}, \mathcal{W}\right) \leq \gamma_{\text{REG}} - \gamma_{\text{OPT}} \leq M^2\left(\mathcal{S}^{\perp}, \mathcal{W}\right). \tag{63}$$

If $v \in \mathcal{S}^{\perp}$ achieves the maximum (minimum) angle with $\mathcal{W}$, then with $x = v + s$ with any $s \in \mathcal{S}$, x achieves

[3]Given two closed subspace $\mathcal{A}_1$ and $\mathcal{A}_2$ of a Hilbert space $\mathcal{H}$, $\cos\left(\mathcal{A}_1, \mathcal{A}_2\right) \triangleq \inf_{a \in C_{\mathcal{A}_1}} \|P_{\mathcal{A}_2} a\|$. Similarly, $M\left(\mathcal{A}_1, \mathcal{A}_2\right) \triangleq \sup_{a \in C_{\mathcal{A}_1}} \|P_{\mathcal{A}_2} a\| = \sqrt{1 - \cos^2\left(\mathcal{A}_1, \mathcal{A}_2^{\perp}\right)} = \sqrt{1 - \cos^2\left(\mathcal{A}_2, \mathcal{A}_1^{\perp}\right)}$.

the upper (lower) bound of (63). Therefore, the bounds of (63) are tight.

An upper bound on the error using the consistent reconstruction method was developed in [1] using the fact that

$$e_{\text{OPT}} = P_{\mathcal{W}^\perp} e_{\text{CON}} \tag{64}$$

and $e_{\text{CON}} \in \mathcal{S}^\perp$ (the latter is easily verified by noting that $x - E_{\mathcal{W}\mathcal{S}^\perp} x = E_{\mathcal{S}^\perp \mathcal{W}} x$). Specifically, it was shown that

$$\|e_{\text{CON}}\| \leq \frac{\|e_{\text{OPT}}\|}{\cos(\mathcal{W}, \mathcal{S})}. \tag{65}$$

Note that since $\cos(\mathcal{W}, \mathcal{S})$ can be arbitrarily close to 0, the reconstruction error can be made arbitrarily large. Using (64) we can also develop a lower bound on the consistent reconstruction error. Since,

$$\frac{\|e_{\text{OPT}}\|}{\|e_{\text{CON}}\|} = \frac{\|P_{\mathcal{W}^\perp} e_{\text{CON}}\|}{\|e_{\text{CON}}\|} \leq M\left(\mathcal{S}^\perp, \mathcal{W}^\perp\right) \tag{66}$$

we have

$$\frac{\|e_{\text{OPT}}\|}{M(\mathcal{S}^\perp, \mathcal{W}^\perp)} \leq \|e_{\text{CON}}\| \tag{67}$$

which, for $M\left(\mathcal{S}^\perp, \mathcal{W}^\perp\right) < 1$, is tighter than the trivial lower bound $\|e_{\text{OPT}}\| \leq \|e_{\text{CON}}\|$ stated in [1]. Combining (65) and (67) we obtain

$$\frac{1}{1 - \cos^2(\mathcal{S}^\perp, \mathcal{W})} \leq \frac{\gamma_{\text{CON}}}{\gamma_{\text{OPT}}} \leq \frac{1}{\cos^2(\mathcal{W}, \mathcal{S})}. \tag{68}$$

As in the case of the bounds (63), it can be shown that the bounds of (68) are tight, by taking $v \in \mathcal{S}^\perp$ which achieves the maximum (minimum) angle with respect to $\mathcal{W}^\perp$ and constructing $x = v + w$ with some $w \in \mathcal{W}$.

6.1 Bound Comparison

We now develop conditions under which the worst-case (best-case) error using the consistent reconstruction is larger than the worst-case (best-case) error using the minimax regret reconstruction.

Using (63) and (68) the worst-case and best-case values of γ_{CON} are larger (respectively) than the worst-case and best-case values of γ_{REG}, only if

$$\frac{\gamma_{\text{OPT}}}{\cos^2(\mathcal{W}, \mathcal{S})} \geq \gamma_{\text{OPT}} + 1 - \cos^2(\mathcal{W}, \mathcal{S}), \tag{69}$$

and

$$\frac{\gamma_{\text{OPT}}}{1 - \cos^2(\mathcal{S}^\perp, \mathcal{W})} \geq \gamma_{\text{OPT}} + \cos^2\left(\mathcal{S}^\perp, \mathcal{W}\right), \tag{70}$$

or, equivalently,

$$\gamma_{\text{OPT}} = \frac{\|P_{\mathcal{W}^\perp} x\|^2}{\|P_{\mathcal{S}^\perp} x\|^2} \geq \cos^2(\mathcal{W}, \mathcal{S}), \tag{71}$$

for the worst case and

$$\gamma_{\text{OPT}} = \frac{\|P_{\mathcal{W}^\perp} x\|^2}{\|P_{\mathcal{S}^\perp} x\|^2} \geq M^2\left(\mathcal{S}^\perp, \mathcal{W}^\perp\right), \tag{72}$$

for the best case. These results are intuitive: If most of the energy of x lies in $\mathcal{S}$ then γ_{OPT} tends to be very large and both inequalities hold. Similarly, if $\mathcal{W}$ and $\mathcal{S}$ are "far apart", then the bounds will be small, and again the inequalities will typically hold. The error resulting from the consistent and regret reconstruction algorithms are illustrated geometrically in Fig. 5. In Fig. 5(a) we depict the consistent and regret reconstruction in the case in which $\mathcal{W}$ is far from $\mathcal{S}$. As can be seen in the figure, in this case the error resulting from the consistent reconstruction is large with respect to the regret reconstruction error. In Fig. 5(b), $\mathcal{W}$ and $\mathcal{S}$ are close, and the errors have roughly the same magnitude.

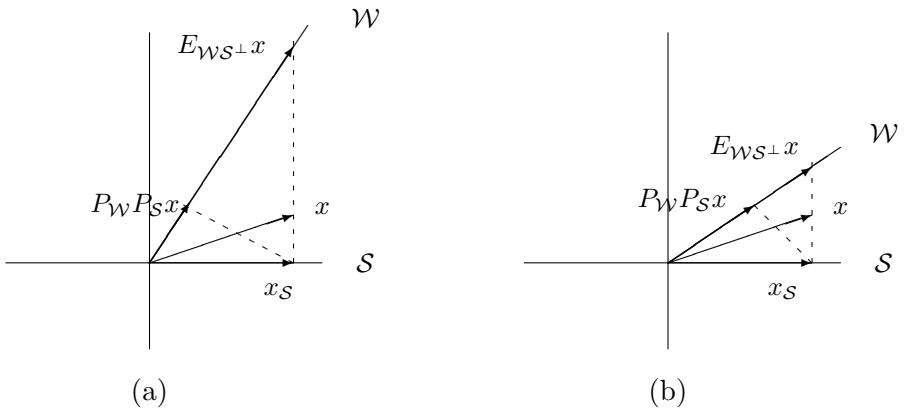


Figure 5: Comparison of the minimax regret reconstruction ($P_{\mathcal{W}} P_{\mathcal{S}} x$) and consistent reconstruction ($E_{\mathcal{W}\mathcal{S}^\perp} x$) for two different choices of $\mathcal{W}$ (a) $\mathcal{W}$ 'far' from $\mathcal{S}$ (b) $\mathcal{W}$ 'close' to $\mathcal{S}$.

Since $\cos^2(\mathcal{W}, \mathcal{S})$ and $M^2\left(\mathcal{S}^\perp, \mathcal{W}^\perp\right)$ are no larger than 1 for all $\mathcal{S}$ and $\mathcal{W}$, it follows that if $\|P_{\mathcal{W}^\perp} x\| \geq \|P_{\mathcal{S}^\perp} x\|$ (or, equivalently, $\|P_{\mathcal{W}} x\| \leq \|P_{\mathcal{S}} x\|$), then both in the best case and in the worst case the minimax regret reconstruction results in a small squared-error than the consistent reconstruction.

Note, that even if (71) and (72) hold, there may exist an x which will yield a smaller squared-norm error using the consistent reconstruction scheme compared with the regret scheme. To ensure that the error resulting from the consistent reconstruction scheme is larger than that resulting from the regret reconstruction scheme *for all* x, we require that the upper bound on γ_{REG} is smaller than the lower bound on γ_{CON}. This condition is fulfilled if

$$\gamma_{\text{OPT}} \geq \frac{M^2\left(\mathcal{S}^\perp, \mathcal{W}\right) M^2\left(\mathcal{S}^\perp, \mathcal{W}^\perp\right)}{\cos^2(\mathcal{S}^\perp, \mathcal{W})} \triangleq \gamma_1. \tag{73}$$

Since the numerator of (73) is no larger than 1, a sufficient condition to ensure a lower error using the regret reconstruction is

$$\gamma_{\text{OPT}} \geq \frac{1}{\cos^2\left(\mathcal{S}^{\perp}, \mathcal{W}\right)}. \tag{74}$$

In a similar manner, we can derive a condition under which the consistent reconstruction scheme results in a smaller error than the regret scheme for all x. Comparing the worst-case bound on the consistent reconstruction error with that of the best-case bound on the regret reconstruction error, we conclude that if

$$\gamma_{\text{OPT}} \leq \frac{\cos^2\left(\mathcal{W}^{\perp}, \mathcal{S}\right)\cos^2\left(\mathcal{W}, \mathcal{S}\right)}{1-\cos^2\left(\mathcal{W}, \mathcal{S}\right)} = \frac{\cos^2\left(\mathcal{S}^{\perp}, \mathcal{W}\right)\cos^2\left(\mathcal{S}^{\perp}, \mathcal{W}^{\perp}\right)}{M^2\left(\mathcal{S}^{\perp}, \mathcal{W}\right)} \triangleq \gamma_2, \tag{75}$$

then the consistent reconstruction scheme will result in a lower error. A sufficient condition is

$$\gamma_{\text{OPT}} \leq \cos^2\left(\mathcal{W}^{\perp}, \mathcal{S}\right)\cos^2\left(\mathcal{W}, \mathcal{S}\right). \tag{76}$$

Since the nominator of γ_2 is no larger than the nominator of γ_1, and the denominator of γ_2 is no smaller than the denominator of γ_1, we conclude that $\gamma_2 \leq \gamma_1$. Thus, in the regime $\gamma_{\text{OPT}} \leq \gamma_2$, we have that $\gamma_{\text{CON}} \leq \gamma_{\text{REG}}$, and in the regime $\gamma_{\text{OPT}} \geq \gamma_1$, we have that $\gamma_{\text{CON}} \geq \gamma_{\text{REG}}$, as illustrated in Fig. 6.

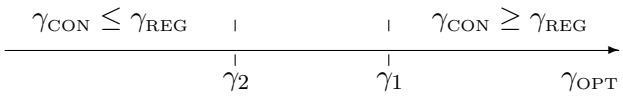


Figure 6: Regions of γ_{OPT} in which the regret reconstruction leads to a smaller error than consistent reconstruction, and vice versa.

7 Examples

We now consider several examples illustrating the minimax regret reconstruction and compare it with the consistent reconstruction method. In Section 7.1 we consider a speech processing example, and in Section 7.2 we consider an image processing example, which also demonstrates the error analysis of Section 6.

7.1 Speech Processing Example

Suppose we sample a continuous-time speech signal $x(t)$ using a non-ideal sampler, so that the samples $c[n]$ are equal to the average of the signal over intervals of length Δ:

$$c[n] = \frac{1}{\Delta}\int_{nT-\Delta}^{nT} x(t)dt. \tag{77}$$

The samples $c[n]$ can be obtained by filtering the signal $x(t)$ with a filter whose impulse response $s(t)$ is given by

$$s(t) = \begin{cases} \frac{1}{\Delta}, & 0 \le t \le \Delta; \\ 0, & \text{otherwise}, \end{cases} \tag{78}$$

and then sampling the output at times $t = nT$. The filter $s(t)$ can be viewed as a (non-ideal) low-pass filter (LPF). In the simulations below, we use $T = 4000^{-1}$s and $\Delta = 1.125$ms. The reconstructed output $\hat{x}(t)$ is obtained from the transformed sequence $d = Hc$ using an interpolation kernel $w(t)$, *i.e.,* $\hat{x}(t) = \sum_n d[n]w(t - nT)$. Specifically, we consider $w(t)$ to be a non-ideal LPF with support on $t \in [0, 1.75]$ms which approximates an ideal LPF with cutoff frequency of 2kHz.

For the purpose of simulation we approximate the continuous-time signal $x(t)$ with a discrete sequence $x[n]$ on a fine grid. In particular, we have used a speech fragment, taken from the Timit database [29], at a sample rate of 8kHz. The continuous time integration kernel $s(t)$ is approximated by the discrete filter

$$s[n] = \begin{cases} \frac{1}{N}, & 0 \le n \le N-1; \\ 0, & \text{otherwise}, \end{cases} \tag{79}$$

with $N = 10$ samples. The ideal sampling is implemented by down-sampling the filter output with a decimation factor of 2. The (non-ideal) LPF followed by decimation can be described by proper construction of the sampling matrix S^*.

To implement the reconstruction we use a linear-phase FIR filter of order 14 (with cutoff frequency 2kHz) as the interpolation kernel. Here as well, the discrete-time interpolation kernel simulates the continuous-time interpolation kernel, by constructing it over the 8kHz fine grid and up-sampling the input sequence d by a factor of 2, prior to filtering. The up-sampling followed by the filtering operation can be described by properly constructing the matrix W.

In Fig. 7 we plot the power spectrum density estimate of the input speech signal and the frequency responses of the sampling and the reconstruction kernels. Fig. 8 presents the input sequence $x[n]$ and 3 different reconstructed signals, corresponding to $H = I$ (that is, not applying a correcting transformation at all), $H = (S^*W)^\dagger$ (consistent reconstruction) and $H = H_{\text{REG}}$ of Theorem 1. As can be seen from the figure, the results of direct reconstruction are poor. This is despite the fact that delay and gain compensations were applied for this method.[4] In particular, the reconstruction filter was multiplied by a factor of 2 as custom in a down-up sampling scheme with a factor of 2. As can be seen from the figure, the consistent reconstruction and minimax regret reconstruction methods perform much better. However, it can be seen

[4]Such compensations are not required by the consistent reconstruction and the regret methods, as they are taken care of automatically by the correction transformation H.

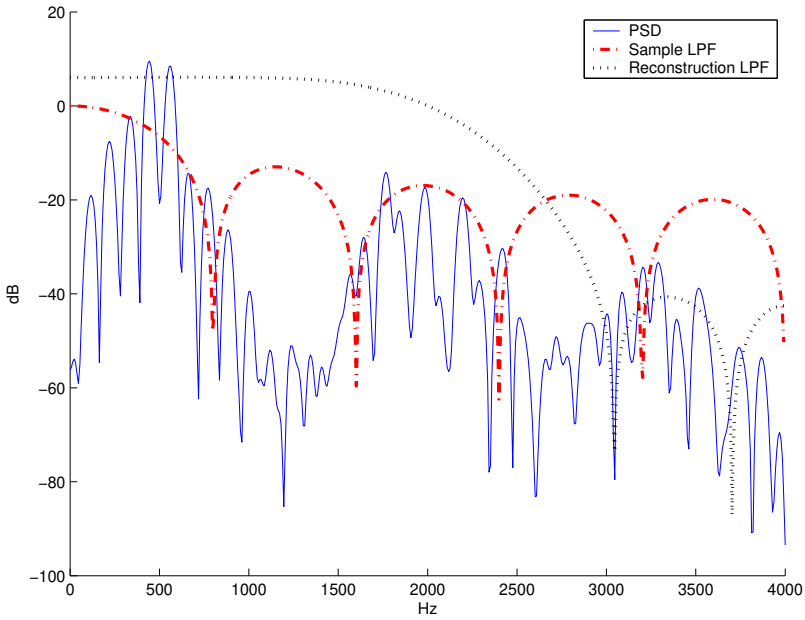


Figure 7: PSD of the speech fragment and the frequency responses of the non ideal low-pass filters.

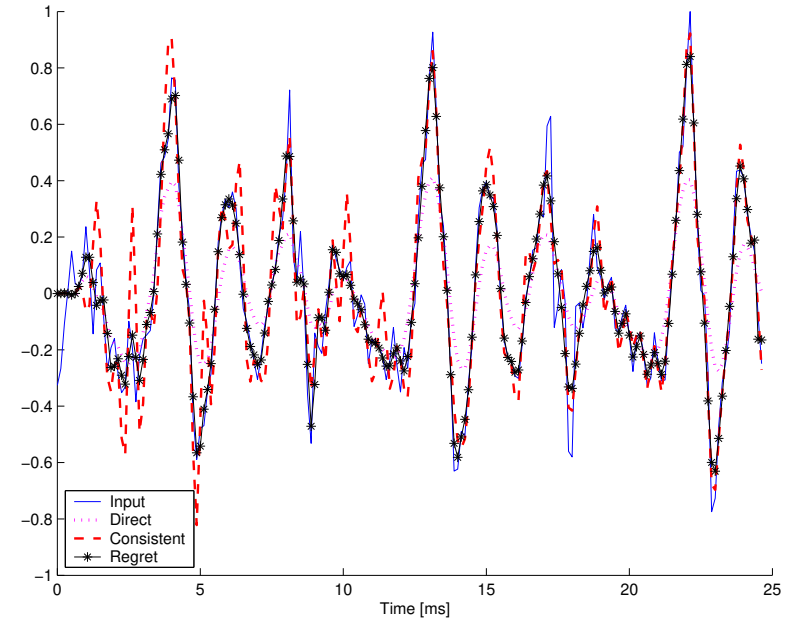
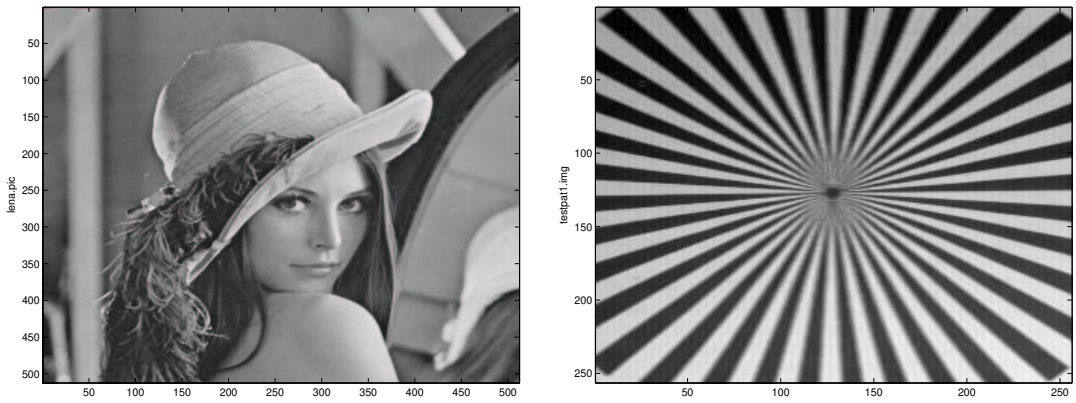


Figure 8: The original speech fragment $x[n]$ and the reconstructed signal using three different methods: direct reconstruction, consistent reconstruction and the minimax regret reconstruction.



Lena image Testpat1 image

Figure 9: The test case images.

that the minimax regret reconstruction leads to better results than the consistent reconstruction method. Computing the error $\|x - \hat{x}\|$ for this example, we obtain that the errors resulting from direct, consistent and minimax regret reconstruction are $2.5977, 1.9111$ and 1.2101, respectively. (The norm of x for this example is 4.47).

Similar results where obtained for other choices of the input signal. Thus, not only does the minimax regret estimator minimize the worst-case error, it seems to lead to superior results for other choices of x as well. We also note that subjective listening tests performed for this setup, confirmed that the minimax regret estimator outperforms the other methods.

7.2 Image Processing Example

We next consider an image processing example, with the goal of partially demonstrating the error analysis of Section 6, in the special case of spline subspaces.

In this example, we consider two possible choices of the input signal x: the 512×512 gray-scale Lena image and the 256×256 gray-scale Testpat1 image, the latter taken from the Matlab® image demos library, both presented in Fig. 9. To obtain a continuous-time representation of these images, we assume that the images are samples of a function $x(t)$ that lies in a subspace $\mathcal{A}$ generated by a spline of order n_a, so that

$$x(t) = \sum_n y[n]\beta^{n_a}(t - n). \tag{80}$$

We further assume that the sampling space $\mathcal{S}$ and the reconstruction space $\mathcal{W}$ are generated by splines of order n_s and n_w, respectively.

As the given image is known only at a discrete grid, we first calculate (within each axis) the spline coefficients $y[n]$ of (80) to obtain a continuous-time representation for the image. These coefficients can be

n_a, n_s, n_w	Direct	Consistent	Regret	Subspace
$1, 0, 1$	846.3	0	533.5	0
$1, 1, 0$	1751.2	1806.6	1711	1711
$2, 1, 0$	2013.3	2079.1	1951.5	1950.8

Table 2: Error norm evaluation (using eq. (83)) for the Lena image, with several different choices of spline spaces.

n_a, n_s, n_w	Direct	Consistent	Regret	Subspace
$1, 0, 1$	820.6	0	525.2	0
$1, 1, 0$	2293.7	2339.8	2271.1	2271.1
$2, 1, 0$	2466.8	2520.2	2435.9	2435.8

Table 3: Error norm evaluation (using eq. (83)) for the Testpat1 image, with several different choices of spline spaces.

computed by direct B-spline filtering [30] with the symmetric IIR filter $\{b^{n_a}\}^{-1}[n]$ as

$$y[n] = \left\{ \{b^{n_a}\}^{-1} * x \right\}[n], \tag{81}$$

where x are the given pixel values. The samples $c = S^* x(t)$ can be obtained directly from the sequence $y[n]$ by noting that

$$c[i] = \langle \beta^{n_s}(t-i), \sum_n y[n] \beta^{n_a}(t-n) \rangle = \sum_n y[n] b^{n_a+n_s+1}[i-n] = \left\{ y * b^{n_a+n_s+1} \right\}[i]. \tag{82}$$

Given the samples $c[n]$, we compute $d = Hc$ using four different choices of H: No correction at all *i.e.,* the direct method $H = I$, consistent reconstruction $H = H_{\text{CON}}$, regret reconstruction $H = H_{\text{REG}}$ and subspace reconstruction $H = H_{\mathcal{A}}$. The reconstructed signal is then given by $\hat{x}(t) = \sum_n d[n] \beta^{n_w}(t-n)$. Under this model, we can compute the reconstruction error in the continuous-time domain using (80) and (58), which results in

$$\begin{aligned} \|\hat{x} - x\|^2 &= \left\| \sum_n d[n] \beta^{n_w}(t-n) - \sum_n y[n] \beta^{n_a}(t-n) \right\|^2 = \\ &= \langle y * b^{2n_a+1}, y \rangle + \langle d * b^{2n_w+1}, d \rangle - 2 \langle y * b^{n_w+n_a+1}, d \rangle. \end{aligned} \tag{83}$$

The error norm of the continuous time representation using the different methods are summarized in Tables 2 and 3 for the Lena image and the Testpat1 image, respectively, using several choices of n_a, n_s and n_w. From Tables 2 and 3 we can see that due to the a-priori knowledge of the subspace in which the input image lies, the subspace method always achieves the lowest reconstruction error possible. When $n_a = n_w$ the consistent reconstruction method (and the subspace method) yield the best reconstruction possible (perfect reconstruction in this case). On the other hand, when $n_a = n_s$, the regret method (and the subspace

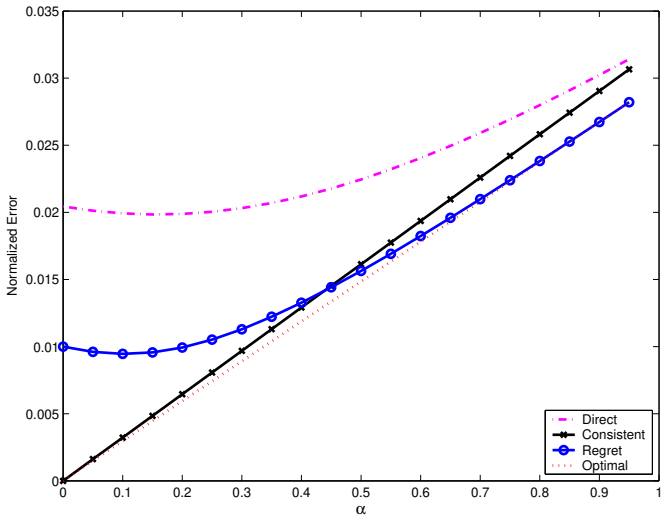


Figure 10: Evaluation of the normalized error $\|x - \hat{x}\| / \|x\|$ for the different reconstruction methods. The relative energy of the input signal within the sampling space (spline of order zero) and the reconstruction space (spline of order 2)is determined by the parameter α.

method) yields the best reconstruction possible. Finally, in the case $n_a = 2, n_s = 1, n_w = 0$ the regret reconstruction is slightly better than the consistent reconstruction. In fact, the regret reconstruction is very close to the optimum, given by the subspace reconstruction.

As our final example, which demonstrates the theoretical analysis of Section 6, we consider the following continuous-time model for the Lena image:

$$x(t) = \alpha \sum_n y_1[n]\beta^{n_s}(t-n) + (1-\alpha)\sum_n y_2[n]\beta^{n_w}(t-n), \tag{84}$$

so that within each axis we model the image as present both in the sampling space and in the reconstruction space, where the parameter $0 \leq \alpha < 1$ determines how 'close' $x(t)$ is to each of these spaces. We note that the suggested convex representation preserves the values of the image on the original grid. In Fig. 10 we plot the normalized error $\|x - \hat{x}\| / \|x\|$ using the different methods (direct (*i.e.,* , $H = I$), consistent, regret, and the optimal least-squares reconstruction). In this example, $n_s = 0$ and $n_w = 2$.

As can be seen from the figure, when α is small, most of the energy of the signal lies in the reconstruction space. Hence, the consistent reconstruction outperforms the regret method. On the other hand, when α is large, most of the energy of the signal lies in the sampling space and the regret method outperforms the consistent reconstruction method. From this figure we can also obtain bounds for γ_1, γ_2 which appear on the right hand sides in equations (73) and (75). Since at $\alpha = 0.45$ (there γ_{OPT} was calculated to be $\gamma_{\text{OPT}} \approx 0.75$) the regret scheme already outperforms the consistent reconstruction scheme we conclude that γ_{OPT} cannot be smaller than γ_2. Hence, for spline subspaces with $n_s = 0, n_w = 2$, γ_2 must be smaller than 0.75. Using similar arguments we can conclude that γ_1 must be larger than 0.5.

8 Conclusion

In this paper, we considered the problem of sampling and reconstruction in arbitrary spaces, using the squared-norm error as the performance measure. If the input signal x lies in an appropriate subspace $\mathcal{A}$ of $\mathcal{H}$, then we showed that a linear reconstruction can be obtained that minimizes the squared-norm error. However, if x is an arbitrary input signal, then the squared-error cannot be minimized using a linear transformation. Therefore, we proposed instead a minimax regret approach that minimizes the worst-case difference between the squared-error and the smallest possible error attainable using a non-linear reconstruction method. We showed that the resulting reconstruction can be interpreted geometrically in terms of the orthogonal projections on the sampling space $\mathcal{S}$ and the reconstruction space $\mathcal{W}$. We also considered efficient implementation of the proposed scheme in the case of SI and spline subspaces.

Finally, we compared the performance of the minimax regret reconstruction with that of the previously proposed consistent reconstruction, and demonstrated both analytically and through simulation, that the minimax regret method can often outperform the consistent reconstruction method.

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